9.8. ELECTRONICS FOR PULSE COUNTING

A wide variety of counting systems has been developed for various purposes. Equipment is often built as NIM (Nuclear Instrument Module standard) or CAMAC (Computer Automated Measurement And Control standard) modules, which fit into standard bins (or crates) containing power supplies and some inter-module connections. Cable connectors are also largely standardized. This facilitates combinations of bias supplies, amplifiers, discriminators, SCAs, ADCs, counters, and other circuitry to fit any need as well as their connection to computers. Several types of counting systems are also designed as plug-in cards to AT-type PC:s. Some non-standard units, e.g. portable instruments, have their own power supply, main amplifier, counter or rate-meter, etc. Today, it is common that even though a detector system comprises all the components above they are not always separate units but integrated in the detector which in turn is connected to a computer using e.g. a USB cable.

9.8.1. Preamplifiers

The purpose of a preamplifier may be twofold. First of all it should increase the energy in the detector signal to such a level that it can drive a reasonable length of low impedance coaxial cable properly terminated at the other end with only a small loss in pulse amplitude. When the detector produces a very weak signal it may be advantageous to amplify that signal to a level where external noise becomes negligible. However, in some applications the detector signal is already large enough, the detector capacitance constant (e.g. a PMT) and *voltage sensitive preamplifiers* with unit gain are used, Figure 9.19(a).

A 0.5 MeV γ absorbed in a germanium crystal will only produce a charge of 2.7×10^{-14} C, Eqn (9.2). Moreover, the detector capacitance may change with operating conditions which make voltage sensitive preamplifiers less suitable and charge sensitive preamplifiers are preferred. The weak charge signal has to be integrated and converted to a voltage pulse by the preamplifier without adding too much noise. Figure 9.19(b) shows a typical input stage for a preamplifier used with germanium γ-ray detectors.

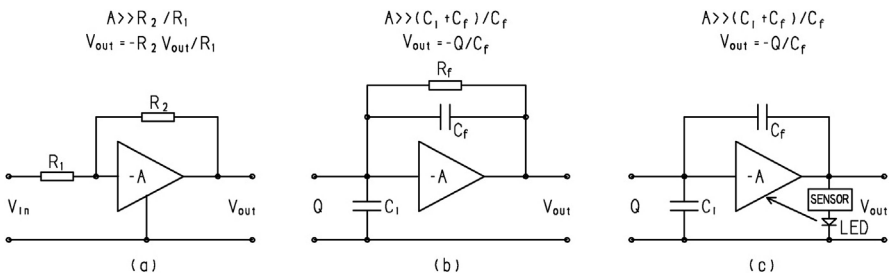


Figure 9.19 Voltage sensitive (a), charge sensitive resistive feedback (b), and pulsed optical feedback (c) preamplifiers. A is amplifier gain.

The function of this stage is to integrate the total charge of the pulse through the detector converting it to a voltage signal with an amplitude that is proportional to the energy deposited in the detector. Often the input FET (Field Effect Transistor) of the preamplifier and the feedback resistor (R_f) are cooled to reduce their contribution to system noise. The charge accumulated on the FET is removed through a resistor − capacitor network, R_f and C_f in Figure 9.19(b), which is part of the amplifier feedback loop. This arrangement is called a *resistor feedback preamplifier*. The feedback resistor adds its inherent Johnson noise to system noise; typically 300 eV in FWHM with a resistor at room temperature. The contribution to the total noise level is, at least for a cooled resistor and FET, negligible in normal γ-spectrometry. However, when measuring low energy X-rays with a Si(Li) detector it may be important to reduce the noise further.

In the *pulsed optical feedback preamplifier*, Figure 9.19(c), the resistor R_f is omitted and the input FET is permitted to charge-up in steps by each new pulse. The output signal is in the form of a staircase function and this is transformed to voltage pulses by differentiating circuits in the main amplifier. When the accumulated charge becomes too large (i.e. the output voltage is near its highest possible value) the sensor circuit triggers and the FET is irradiated with a short light pulse from the built-in light emitting diode thereby causing a short circuit in the FET, removing the accumulated charge. The reset operation produces a large voltage swing at the preamplifier output which may easily overload the main amplifier and make it distort pulses rapidly following the reset pulse. In order to avoid making measurements during and immediately after the big reset pulse a blocking signal is often available from the preamplifier during the reset pulse. A typical pulsed optical feedback preamplifier can, with a Si(Li) X-ray detector, give a FWHM of <150 eV at 5.9 keV. The good resolution is important in low energy applications, e.g. X-ray measurements. A minor drawback of pulsed optical feedback is the extra dead time introduced by blocking measurements during each reset operation.

The output from a voltage sensitive or resistor feedback preamplifier is a tail pulse with a rather long decay time. Hence, some pulse pile-up is unavoidable, except at very low count rates. Pile-up will cause the average level of this signal to increase with pulse rate, which may approach the limit of linear operation of the preamplifier.

9.8.2. Amplifiers

The purpose of an amplifier to amplify a voltage pulse in a linear fashion and to shape the pulse so that the event can be analyzed easily and correctly in a short time. A linear amplifier accepts tail pulses as input, usually of either polarity, and produces a shaped and amplified pulse with standard polarity and amplitude span (NIM standard is positive polarity and 0 to 10 V amplitude). On most commercial linear amplifiers, the time constants for the various pulse shaping circuits are adjustable to fit various detector and count-rate requirements.

Biased amplifiers shifts the zero of the amplified pulse down while still producing only positive output pulses. This effect is important in e.g. α-spectrometry as the lower energy range is normally of no interest and it is better to use the available output amplitude span to enlarge the interesting high energy region than to cover the whole energy range.

Amplifiers with a logarithmic amplitude response have use in liquid scintillation counting for compression of the higher energy part of the β-spectrum.

9.8.3. Single channel analyzers

The pulse obtained from many detectors is directly proportional to the energy absorbed in the detector. If all of the energy of the nuclear particle is absorbed in the detector, which is possible for all kinds of ionizing radiation, and only one particle or photon interacts at a time with the detector, the energy of the particle or photon can be determined from the size of the generated pulse. In single-channel analyzers (SCA) only one channel exists which serves as a "window" to accept only pulses of a certain size corresponding to a limited range of energy as indicated in Figure 9.5(b). This window can be moved in steps through the entire energy range, thereby obtaining a measure of the count rate of particles having different energies. In the Figure the window position and width are set so as to cover only the middle peak. By narrowing the window, and moving it from zero to maximum pulse amplitude, the whole particle energy spectrum is obtained. Spectrometry represents one of the principal techniques whereby the energy of ionizing radiation is measured. The single channel spectrometer is a system for energy determinations, although multichannel systems are simpler to use, more accurate and faster.

The main use of SCAs is to select events within a given energy interval. Thereby the background count rate is very much reduced which increases sensitivity. One of the most important routine uses of SCAs is in simultaneous ^{3}H and ^{14}C counting using liquid scintillators. Two SCAs are normally used in, one set to cover pulses corresponding to the very soft β^- from ^{3}H and the other set to cover the more energetic pulses caused by the β^- from ^{14}C.

9.8.4. Counters and rate meters

Ultimately the amplified pulses must often be accumulated over a fixed time interval and the resulting number displayed. The device used is based on a simple electronic register which is incremented by one each time a pulse arrives at its input. Such devices are often called *counters* or *scalers*. Counters can usually be operated in two modes, *preset time* or *preset count*. Counting is started and stopped by manual push buttons or by logic signals from other units, e.g. the logic stop signal from a timer in the preset time mode. In the preset count mode, the scaler stops at a given count and produces a logic output signal which can be used by other units, e.g. to stop a timer.

Multi-channel scalers are devices where the counts are accumulated in memory cells similar to a MCA, cf. section 9.8.5. In operation counts are accumulated in a current memory cell and the cell is changed to the next higher cell after a preset time interval. These devices are often in the shape of a circuit board for use in a PC. Start, stop, channel change, reset, and readout are then performed through software running on the PC.

In some applications a continuous display or record of the count rate is desired, e.g. in a survey meter. This is achieved by use of a *count rate meter*. Count rate meters can be analog (based on a diode pump) or digital (based on a recycling scaler-timer-display unit or a multi-channel scaler with preset dwell time for the channels). In either case, the value obtained follows the input rate with a certain time-lag, determined by the *time constant* used. A high time constant gives a smooth reading as it averages input pulse rate fluctuations over a longer time interval, but a very sluggish response to true input rate changes. Correspondingly a short time constant gives a rapid response, but also a very fluctuating reading.

9.8.5. Multichannel analyzers

Multichannel analyzers (MCA) consisting of an analog to digital converter (ADC), controller and storage unit which may have 16 000 channels or more (i.e. the energy scale is split up into that number of steps). In this case the pulses are sorted immediately into the various channels which record the counts as they occur rather than scan over an energy range in steps. In many designs the MCA is interfaced to a computer to provide display, recording and analysis of the energy spectrum. MCAs are also made as circuit boards that fit inside a personal computer and have suitable software by which the PC can emulate a stand-alone dedicated MCA. The ADC unit is normally based on one of two principles.

In a Wilkinson ADC, the beginning of an input pulse starts a pulser (the clock) and a ramp voltage. When the input signal culminates, its amplitude is "frozen" by a sample-and-hold gate. The pulser runs until the ramp voltage crosses the frozen input signal level. By accumulating the pulser signals in a counter the final count is proportional to the amplitude of the input pulse. This count can then be used as a digital address, *channel*, telling the attached digital circuitry where to increment the stored counts. The dead time of this ADC type is roughly proportional to the amplitude of the input pulse and to the highest address permitted. A typical conversion time with a 100 MHz clock may be $1 + 0.01n$ μs, for channel n. Wilkinson ADCs are mostly used in MCAs with a relatively small number of channels.

The successive approximation ADC also locks the maximum amplitude in a sample-and-hold circuit and then uses an ultra fast digital voltmeter of successive approximation type to measure the amplitude. A typical conversion time is ~25 μs at 4000 channels full scale. The voltmeter reading is then used as address in the same way as for the Wilkinson ADC. This type of ADC has usually a fixed dead time, independent of the pulse amplitude

and only moderately dependent on the maximum address (each doubling of the total number of channels adds the same increment to the conversion time). Successive approximation ADCs are preferred for MCAs with very many channels as they then, on the average, become faster than Wilkinson ADCs. On the other hand the linearity of a Wilkinson ADC is usually slightly better than for a successive approximation ADC.

MCAs are probably the most versatile instruments for nuclear particle detection because of their usefulness both for qualitative identification and quantitative determination of radioactive nuclei. Practically all α- and γ-spectra reproduced in this book have been obtained through this technique.

In case correlated events registered by several ADCs are measured the addresses from the ADCs and time is usually recorded digitally on a magnetic storage medium in real time. The desired information, e.g. $\gamma-\gamma$ coincidences, is then obtained by reading and sorting the stored data after the experiment.

9.8.6. γ-spectrometry

As we learned in section 7.5, the capture of a γ-ray in an absorber such as a NaI(Tl)- or Ge-crystal occur by any of three processes – photoelectric effect, Compton effect, and pair production. In energy analysis of γ-rays it is desirable to capture the total energy and to minimize the loss of energy by escape of the scattered γ-rays from Compton interaction. This increases the number of events contributing to the photo peak which corresponds to the total γ-ray energy. Also, if lower energy γ-rays are present their photopeaks may be obscured by the Compton distribution from higher energy γ-rays. With anthracene crystals little resolution is seen in γ-spectra as the low atomic number of the absorber makes the principal interaction the Compton effect. However, in NaI(Tl), CsI(Tl), Si(Li), Ge(Li) and HPGe crystals the photoelectric effect is the more important. Increasing the size of the crystal increases the probability of photon capture and, therefore, the probability of capture of the scattered γ-ray in Compton events is increased. As a result, the increase in crystal size results in more capture of the total incident γ-ray energy which appears under the photo peaks.

For an ideal infinitely small detector a γ-spectrum from ^{137}Cs would have the shape shown in Figure 9.20(a). For comparison theoretical γ-spectra for finite size and infinitely large detectors are shown in the same Figure. The counts located between the compton edge and photo peak for the medium size detector are due to multiple Compton events. As can be seen detector size has a big influence of the Compton continuum and photo peak. A measured spectrum is shown in Figure 9.20(b). The energy of the Compton edge, E_{CE} (keV), caused by a γ-line of energy E_γ (keV), is given by

$$E_{CE} = E_\gamma / \left[1 + 511/\left(2E_\gamma\right)\right] \tag{9.29}$$

where 511 is the energy in keV equivalent of one electron mass.

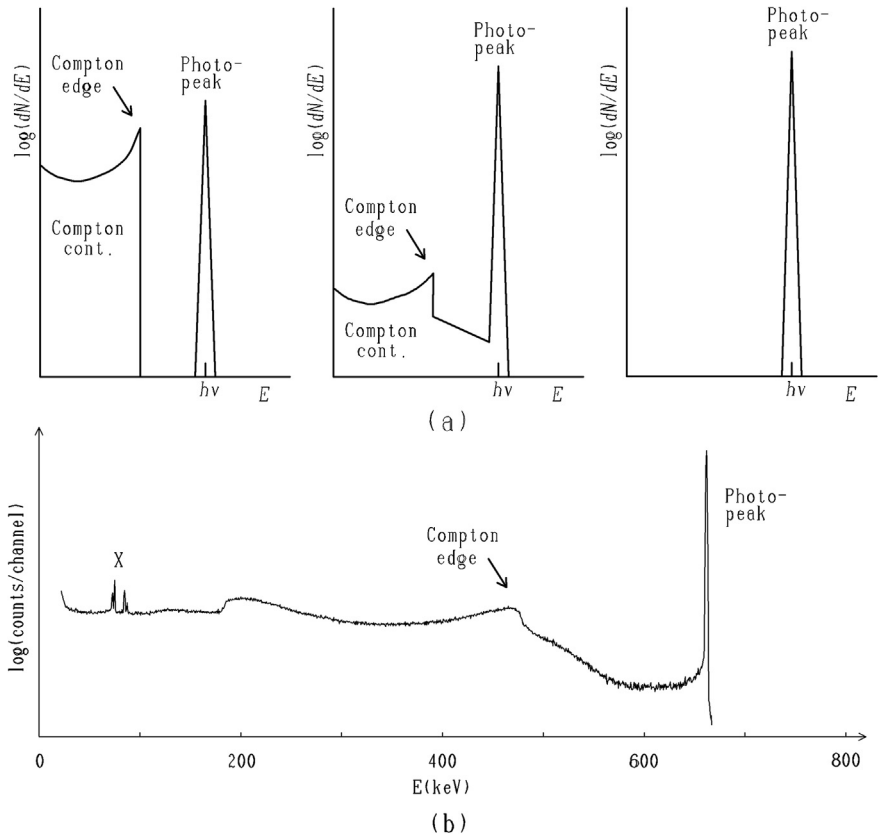


Figure 9.20 (a) ^{137}Cs γ-spectrum as calculated for "ideal" very small, medium and infinite size γ-detectors (b) the spectrum really recorded with a 22% HPGe detector.

The broadening of the photo peak has many causes such as inhomogeneities in the crystals and variations in charge, or light, collection and noise in the preamplifier. However, for scintillation detectors the main cause is found in the PMT where non-uniformity in the photocathode, fluctuations in the high voltage imposed on each dynode, and statistical variations in the small number of photoelectrons formed at the photocathode are all contributing factors.

The resolution is the determining factor in the ability of the system to differentiate between photopeaks of γ-rays of similar energy. Other features of the spectrum in Figure 9.20(b) are the backscatter peak and the X-ray peaks. The broad backscatter peak, located between the X-rays and the compton edge, arises by the absorption in the crystal of scattered photons resulting from γ-ray absorption via Compton interactions in the material surrounding the crystal. Obviously the magnitude of this peak is dependent on the distance of this material from the crystal and on the nature and amount of the

material. The X-ray peaks (X in Figures 9.16 and 9.20(b)) are due to the absorption of the X-rays emitted in the electronic rearrangement following the nuclear disintegration, following internal conversion or after excitation of materials near the detector. Bursts of very low energy electrons are generated thermally in a semiconductor or emitted spontaneously from the photocathode in a PMT and are the cause of a noise peak at very low discriminator settings. This often limits the photon energies that can be studied to a minimum of some keV.

For γ-energies above 1.02 MeV, pair production in the detector followed by annihilation of the positron leads to the generation of pairs of 0.511 MeV photons. The probability increases rapidly with energy above 1.02 MeV. One or both of these photons may escape from the detector giving a deposited energy corresponding to the initial γ-energy less one or two times 0.511 MeV. Positrons are also generated outside the detector and their annihilation radiation reaches the detector. This gives rise to three smaller photo peaks called *escape peak*, *annihilation peak* and *double escape peak* at the corresponding energies, $E_\gamma - 0.511$ MeV, 0.511 MeV and $E_\gamma - 2 \times 0.511$ MeV, marked 1e, A and 2e respectively in Figure 9.16.

A complicating factor in gamma spectroscopy is the possibility of summing of two or more different gammas into one peak, so called coincident summing. This effect can originate from two ways: random coincidence and true coincidence summing (TCS). Random coincidence is the result of two separate gamma photons from different decays hitting the detector at the same time, i.e. within the resolution time of the detector. The energy of these gammas will then be summed and registered as a single event. This effect is dependent on the actual activity of the sample. The higher the activity the more probable that two or more decays occur simultaneously. In the second case, true coincedence summation, the origin is from a complex decay scheme, i.e. for each original decay there are several gammas emitted more or less simultaneously. This is not activity dependent and can cause severe calibration problems if not taken into account. This phenomenon is sometimes also called cascade summing. In more detail one may say that coincident γ-rays emitted in rapid cascade during one decay may hit the detector at the same time and register as one event with the combined energy. This gives two effects, firstly the generation of a *sum peak* containing one count per coincidence and secondly corresponding loss of one count in each of the peaks corresponding to the coincident photons. Summing is strongly dependent on detector size and source to detector distance.

The photo peaks from a well behaved detector are almost Gaussians. Thus the FWHM value determines the peak shape. The energy scale is calibrated against standard isotopes and is normally almost linear. In most applications only the area below the photo peak but above the underlying continuum is used as a measure of activity. The total efficiency (based on photo peak area) varies usually with energy in the way illustrated by Figure 9.21.

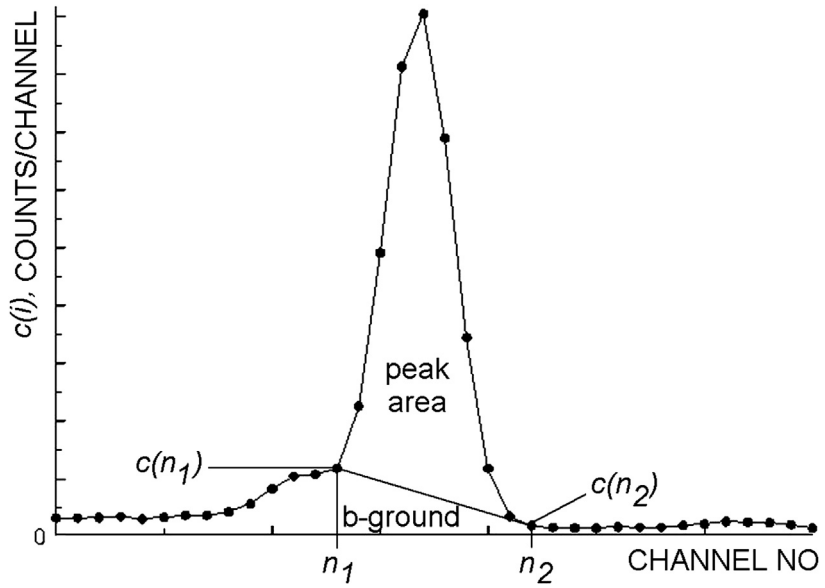


Figure 9.21 Typical efficiency curves for HPGe detectors. GEM and GMX are detectors of n- and p-type with Be-window respectively. (Acc. to ORTEC.)

The determination of a net peak area in α, γ, and X-ray spectroscopy requires a calculation and subtraction of the background in the peak region. Figure 9.22 shows a typical peak which begins in channel n_1 and ends in channel n_2. If no other information about the shape of the background is available it is usually assumed to be a linear function

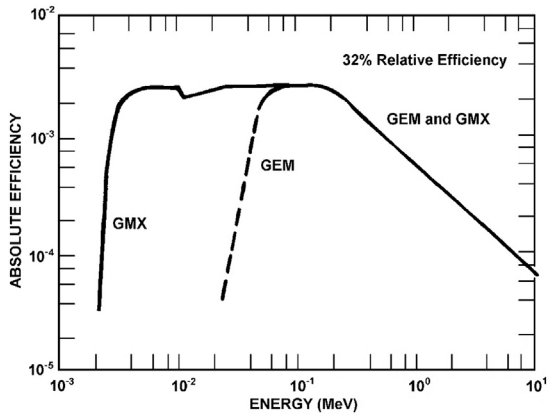


Figure 9.22 A typical peak from a HPGe detector and notations used in eqns. (9.30) and (9.31) for calculation of the net peak area and its standard deviation.

of the channel number between the two limiting channels, n_1 and n_2. The net peak area count, N_{peak}, is then calculated from

$$N_{\text{peak}} = \left[\sum_{i=n_1}^{n_2} c(i)\right] - 0.5 \times (n_2 - n_1) \times [c(n_1) + c(n_2)] \tag{9.30}$$

where $c(i)$ is the count in channel i. The uncertainty in N_{peak}, $s_{N_{peak}}$, can be estimated from

$$s_{N_{peak}} = \sqrt{\left[\sum_{i=n_1}^{n_2} c(i)\right] + 0.5 \times (n_2 - n_1) \times [c(n_1) + c(n_2)]} \tag{9.31}$$

9.9. SPECIAL COUNTING SYSTEMS

For *low intensity measurements*, in which a level of radioactivity comparable to the normal background radiation is to be measured, special electronic circuits incorporating two detectors are used. The detectors are coupled so that a signal registers only when both detectors are activated at the same time (*coincidence circuit*) or, alternatively, when only one but not the other is activated at the same time (*anticoincidence circuit*). With such arrangements the normal β, γ background of a detector may be decreased by more than a factor of 100. In the most advanced coincidence techniques both detectors are energy sensitive as well, providing information on the type of radiation being measured. For example, β-γ coincidence measurements are used for absolute determination of radioactivity for samples in which β-decay is immediately followed by γ-emission.

Detectors which are *direction sensitive* have been developed primarily for use in medical diagnosis with radioactive isotopes. The simplest version involves a scintillation detector surrounded by a lead shield with a small hole (collimator) through which radiation reaches the detector. More complex systems with many detector/collimator pairs are in common use (γ-cameras); cf. section 18.5. Scanning instruments (Fig. 18.13) have been developed which permit the measurement of radioactivity as a function of several coordinates as illustrated in Figure 18.14. Such instruments make it easy to detect the accumulation of a radioactive tracer in a particular organ of the body.

Whole-body counters were originally developed for investigation of poisoning by radioactive substances such as radium. They are now used diagnostically and consist of a large scintillation, or semiconductor, detector with the whole system, including the patient, placed in a heavily shielded room. The sensitivity is sufficient to measure natural radioactivity in the human body from such nuclides as ^{40}K.

In nuclear installations like uranium mines, nuclear reactors, reprocessing plants, etc., it is necessary to continuously monitor gas and liquid effluents. Figure 9.23 shows an